

AI-Generated Image Detection

Using Frozen CLIP Features and Linear Probing

Re-implementation of UniversalFakeDetect (Ojha et al., CVPR 2023)

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Final Results Summary

PModel	PAUC	PAccuracy	PF1	PAP
Random Classifier	0.502	0.502	0.500	0.498
Simple CNN (baseline)	0.973	0.901	0.904	0.973
CLIP Linear Probe (Ours)	0.9247	0.8349	0.8456	0.9283

GitHub: <https://github.com/hegoo2004/CSE429-AI-Image-Detection>

Live Demo:

<https://huggingface.co/spaces/hego2004/ai-image-detector>

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The rapid advancement of generative models has made AI-generated image detection a critical challenge for digital forensics and media integrity. Prior methods suffer from poor cross-generator generalisation. This project re-implements **UniversalFakeDetect** (Ojha et al., CVPR 2023), performing binary real-vs-fake classification in the feature space of a frozen CLIP-ViT-L/14 encoder. Training only **769 parameters**, our model achieves **AUC = 0.9247**, **F1 = 0.8456**, and **AP = 0.9283** on a 3,600-image test set from two generator families. The model is deployed as a live web application on Hugging Face Spaces.

1. Introduction

The proliferation of AI-generated imagery from GANs, DDPMs, and LDMs has fundamentally altered digital media. State-of-the-art generators produce photorealistic images indistinguishable from real photographs by human observers, creating serious risks for misinformation, identity fraud, and political manipulation.

The principal challenge in detection is **cross-generator generalisation**: a classifier trained on ProGAN images achieves near-perfect accuracy on ProGAN test images but collapses to near-chance performance on Stable Diffusion or DALL-E outputs. This project implements **UniversalFakeDetect**, which directly addresses this limitation by classifying images in the feature space of a frozen CLIP-ViT-L/14 encoder.

1.1 Motivation

- AI-generated images are increasingly weaponised for disinformation
- Prior detectors generalise poorly across generator families
- A lightweight universal solution is needed for practical deployment

2. Related Work

2.1 Prior Detection Methods

CNNDetection (Wang et al., CVPR 2020) fine-tunes ResNet-50 on ProGAN images with aggressive augmentation. Strong within-family generalisation but $\text{AUC} \approx 0.51$ on diffusion outputs.

GramNet (Liu et al., CVPR 2020) augments ResNet with Gram matrix blocks. Improves GAN detection but fails similarly on diffusion outputs.

LNP (Liu et al., ECCV 2022) exploits the absence of natural camera noise in AI images, achieving better cross-family performance but degrading on post-processed images.

Method	Backbone	PGAN	pDiffusion	pParams
CNNDetection	ResNet-50	Strong	~3% AUC	25M
GramNet	ResNet+Gram	Good	Limited	28M
LNP	Lightweight CNN	Moderate	Moderate	5M
UniversalFakeDetect	CLIP-ViT-L/14	Strong	Strong	769

Sink Class Phenomenon: Prior methods collapse the fake class to a tight cluster in feature space. Unseen generators fall outside this cluster and are misclassified as real. Frozen CLIP features avoid this failure entirely.

3. Approach

3.1 Architecture

Our **LinearProbingDetector** consists of two components:

- 1. **Frozen CLIP-ViT-L/14 Backbone:** Produces a 768-dim L2-normalised feature vector. All 307,302,976 parameters are frozen.
- 2. **Trainable Linear Head:** Single Linear(768 → 1) with bias. Only **769 parameters** are trained.

$$\hat{y} = \sigma(W \cdot f_{\text{CLIP}}(x) + b), \quad W \in \mathbb{R}^{1 \times 768}, b \in \mathbb{R} \tag{1}$$

Module	Output Shape
Input Image	Preprocessed RGB [B, 3, 224, 224]
CLIP Visual Enc.	ViT-L/14 (frozen, 307M) [B, 768] (L2-norm)
Dropout	p = 0.2 [B, 768]
Linear Head	Linear(768 → 1) + bias [B, 1] (logit)
Output	$\sigma(\text{logit}) \geq 0.4 \Rightarrow \text{Fake}$ [B] ∈ {0, 1}

Component	Parameters	Trainable?	Source
CLIP ViT-L/14 backbone	307,302,976	No (frozen)	OpenAI CLIP
Linear head	769	Yes	Xavier init
Total	307,303,745	769 (0.0003%)	—

3.2 Training Configuration

- Optimiser: AdamW (lr=0.001)
 - Weight decay: 0.001
 - Scheduler: Cosine Annealing
 - $T_{\max} = 20, \eta_{\min} = 10^{-5}$
 - Loss: BCEWithLogitsLoss
- Dropout: $p = 0.2$
 - Gradient clipping: max 1.0
 - Early stopping: patience=5
 - Batch size: 32
 - Hardware: RTX 3050 6GB GPU

4. Datasets

Dataset	Split	Total	Real	Fake	Generator
CIFAKE	Train	7,000	3,500	3,500	Stable Diffusion v1.4
	Val	1,500	750	750	
	Test	1,500	750	750	
Real & Fake Face	Train	7,000	3,500	3,500	StyleGAN
	Val	1,500	750	750	
	Test	1,500	750	750	
Combined	Train	14,000	7,000	7,000	Mixed
	Test	3,000	1,500	1,500	

Preprocessing: All images resized to 224×224 (Lanczos), converted to RGB, normalised with ImageNet statistics. Training augmentation: RandomHorizontalFlip + ColorJitter. WeightedRandomSampler ensures per-batch class balance.

5. Experiments & Results

5.1 Quantitative Results

PModel	PAUC	PAccuracy	PF1	PAF
Random Classifier	0.502	0.502	0.500	0.498
Simple CNN (4-layer, scratch)	0.973	0.901	0.904	0.973
CLIP Linear Probe (Ours)	0.9247	0.8349	0.8456	0.9283

Our model achieves **AUC = 0.9247** training only **769 parameters** — a 0.4228 improvement over chance. The Simple CNN’s higher same-distribution accuracy reflects overfitting; our approach is designed for cross-generator generalisation which the CNN cannot provide.

5.2 Confusion Matrix

	Predicted Real	Predicted Fake
True Real	TN = 2,188 (76.1%)	FP = 687 (23.9%)
True Fake	FN = 256 (9.0%)	TP = 2,582 (91.0%)

The model achieves **91.0% fake recall** vs **76.1% real recall**. The asymmetry reflects greater diversity in real images — CIFAKE real images (upscaled CIFAR-10) differ significantly from natural photographs.

5.3 Training Dynamics

Both loss curves decrease smoothly over 20 epochs. The train-validation accuracy gap stabilises at **2–3%**, confirming well-controlled overfitting.

- Epoch 1: val_acc = 0.738, val_auc = 0.854
- Epoch 10: val_acc = 0.818, val_auc = 0.912
- Epoch 20: val_acc = 0.835, val_auc = **0.925** (best checkpoint)

5.4 Score Distribution

P(fake) score analysis on the test set:

- **Real images:** Broad distribution concentrated near 0
- **Fake images:** Sharp peak near 1.0 indicating high-confidence detection
- **Overlap region** [0.3, 0.7]: Genuinely ambiguous boundary cases

6. Qualitative Results

6.1 Success Cases

- **StyleGAN faces:** $P(\text{fake}) > 0.90$ — smooth textures and unnatural eye reflections reliably detected
- **Stable Diffusion images:** $P(\text{fake}) > 0.85$ — diffusion frequency artifacts identified
- **Natural textured photographs:** Correctly classified as real with high confidence

6.2 Failure Cases

- **Modern smartphone photos:** Classified as fake due to distributional mismatch with training real images
- **HDR/high-saturation photographs:** Post-processing pushes embeddings toward fake region
- **High-quality AI images:** Photorealism-optimised generators reduce the feature-space gap

Root Cause: Training on two datasets limits real image diversity. Adding COCO or ImageNet real images would significantly improve real-photo detection from cameras and phones.

7. Deployment

Live Demo:

<https://huggingface.co/spaces/hego2004/ai-image-detector>

GitHub: <https://github.com/hegoo2004/CSE429-AI-Image-Detection>

The model is deployed as a **Gradio web application** on Hugging Face Spaces (free tier):

- CLIP backbone downloaded automatically from OpenAI on first run
- Only the 769-parameter linear head stored in `best.pt` (5 KB)
- User uploads image → CLIP features → linear score → Real/Fake verdict
- Response includes label, confidence percentage, and $P(\text{fake})$ score

8. Conclusion & Future Work

We re-implemented UniversalFakeDetect, demonstrating that a frozen CLIP-ViT-L/14 backbone with only 769 trainable parameters achieves **AUC = 0.9247** and **AP = 0.9283**. Training curves confirm stable convergence with no significant overfitting, and the confusion matrix reveals strong fake detection recall (91.0%).

Future Work:

- Add COCO/ImageNet real images to improve real-photo detection
- Zero-shot evaluation on DALL-E 3, SDXL, and Midjourney v6
- Selective fine-tuning of later ViT layers without losing generalisation
- Adversarial robustness evaluation against CLIP-targeted attacks

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